Chittagong University of Engineering & Technology

Department of Electronics and Telecommunication Engineering

EXPERIMENT NO: 06

Name of the Experiment

ACL configuration and traffic filtering using CISCO packet tracer

Course No : ETE 412

Course Title : Advanced Communications and Networking Sessional

Date of Experiment : 11/01/2026

Date of Submission : 25/01/2026

REMARKS

Objectives

• To configure Access Control Lists (ACL) for traffic filtering in a network.
• To control network access based on host and network-level conditions.
• To observe packet filtering behavior using ACL rules.

Software

• Cisco Packet Tracer

Introduction

Access Control Lists (ACLs) are used to filter network traffic and provide security by permitting or denying packets based on defined rules. Standard ACLs filter traffic based on source IP address, while extended ACLs consider source, destination, and protocol. ACLs are applied on router interfaces to control incoming or outgoing traffic.

Procedure

1. Network topology was created using routers, switches, and PCs.
2. IP addresses and subnet masks were assigned to all devices.
3. Default gateways were configured on all PCs.
4. Router interfaces were activated using `no shutdown`.
5. ACLs were created using `access-list` commands.
6. ACLs were applied to router interfaces using `ip access-group`.
7. Traffic filtering was tested using ping between selected hosts.

Connection Diagram

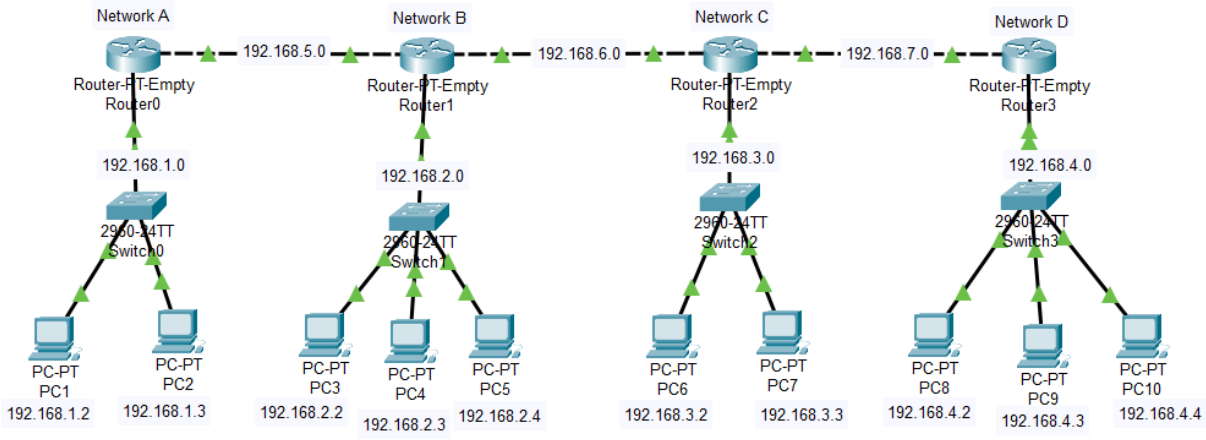


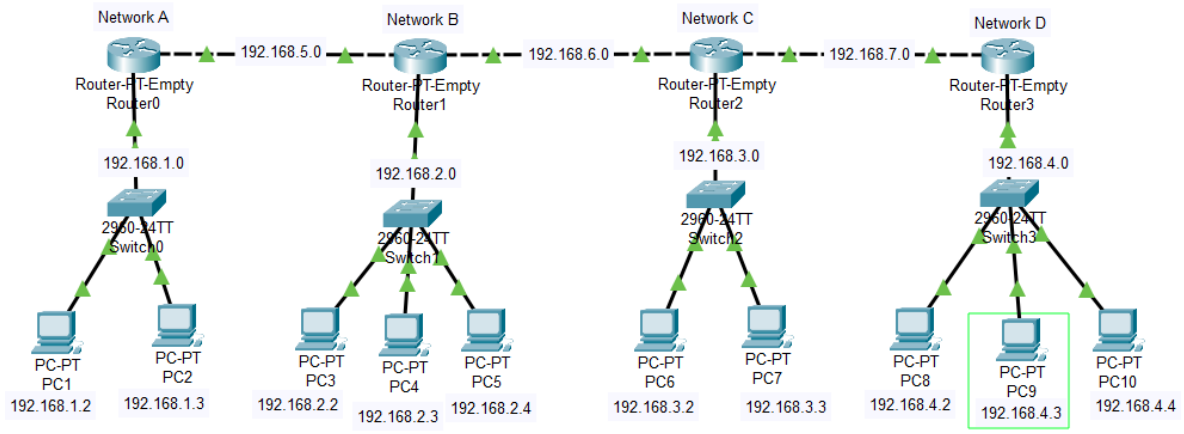
Figure 6.1 : ACL configuration and traffic filtering using multiple routers and LANs.All networks were configured with subnet mask **255.255.255.0**

Segment Type	Network Name	Network Address	Default Gateway
LAN	Network A	192.168.1.0	192.168.1.1
LAN	Network B	192.168.2.0	192.168.2.1
LAN	Network C	192.168.3.0	192.168.3.1
LAN	Network D	192.168.4.0	192.168.4.1
WAN Link	R0–R1	192.168.5.0	Serial Link
WAN Link	R1–R2	192.168.6.0	Serial Link
WAN Link	R2–R3	192.168.7.0	Serial Link

Table 6.1 : Network Addressing Table for the figure 1 Topology.

ACL Configuration Tasks

Task 1: Permit Host PC9 and Deny All Others



Only PC9 (e.g., **192.168.4.3**) is allowed

Commands:

```
enable
conf t
access-list 10 permit host 192.168.4.3
access-list 10 deny any
interface Gig1/0
ip access-group 10 in
end
```

Output

PC6 Before Task 1 ACL rules

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=125
Reply from 192.168.1.2: bytes=32 time<1ms TTL=125
Reply from 192.168.1.2: bytes=32 time=1ms TTL=125
Reply from 192.168.1.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC9 After Task 1 ACL rules allows only PC9

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=124
Reply from 192.168.1.2: bytes=32 time<1ms TTL=124
Reply from 192.168.1.2: bytes=32 time=1ms TTL=124
Reply from 192.168.1.2: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC6 After Task 1 ACL rules blocks others (Here PC6)

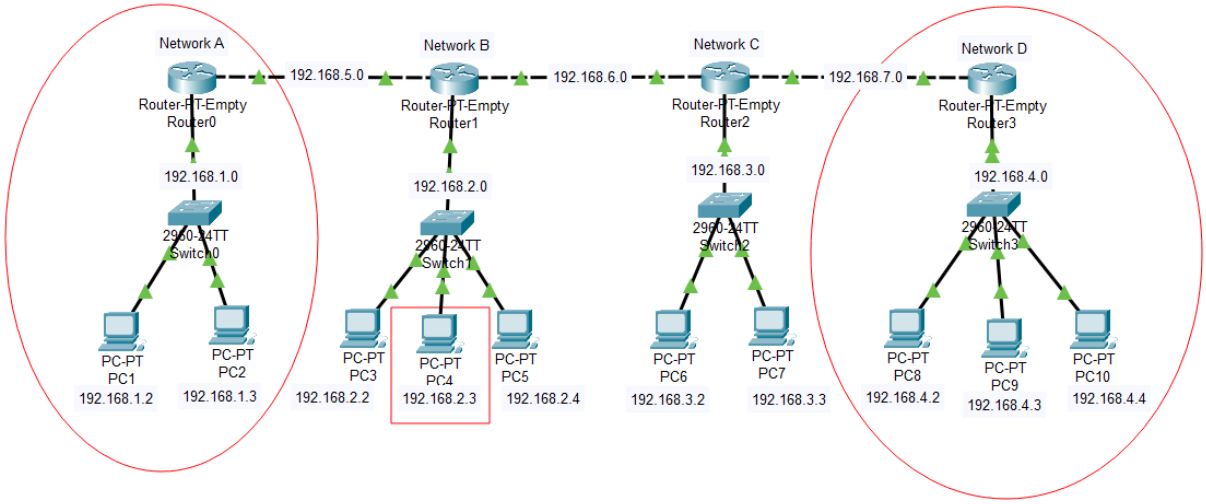
```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.5.2: Destination host unreachable.
Reply from 192.168.5.2: Destination host unreachable.
Reply from 192.168.5.2: Destination host unreachable.
Reply from 192.168.5.2: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 2: Remove Previous ACL and Deny Host PC4 to Network A & D, Permit Others



Step 1: Remove old ACL

Run on routers A where ACL was applied before:

```
enable
conf t
no access-list 10
no access-list 20
```

Step 2: Create ACL (on both Router0 & Router3)

```
access-list 20 deny host 192.168.2.3
access-list 20 permit any
```

Output

PC4 Pinging to network A Fails

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.5.2: Destination host unreachable.
Reply from 192.168.5.2: Destination host unreachable.
Reply from 192.168.5.2: Destination host unreachable.
Reply from 192.168.5.2: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

PC4 Pinging to network D Fails

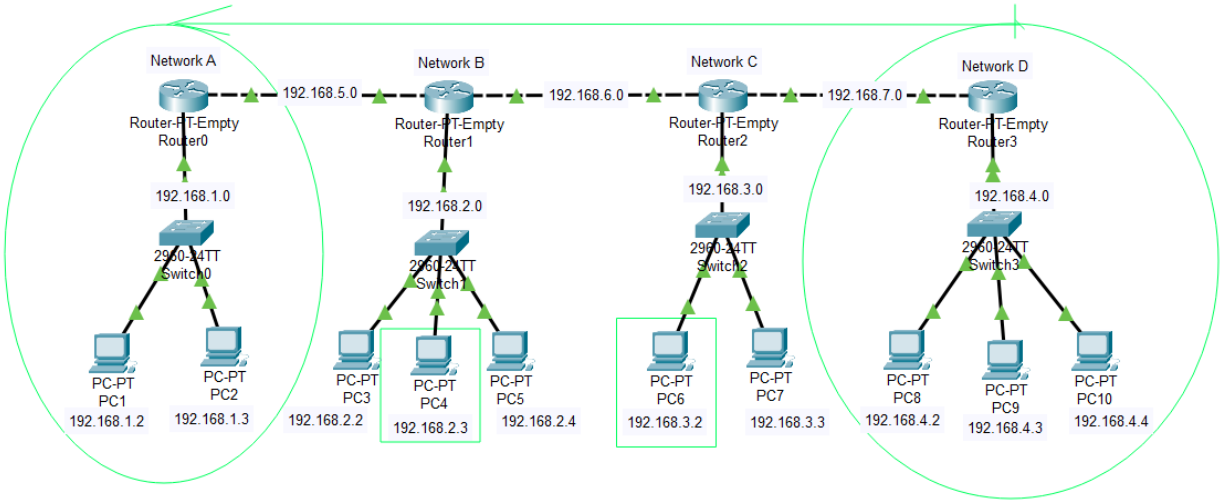
```
C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Reply from 192.168.7.1: Destination host unreachable.
Reply from 192.168.7.1: Destination host unreachable.
Reply from 192.168.7.1: Destination host unreachable.
Reply from 192.168.7.1: Destination host unreachable.

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 3: Allow Network D to Access Network A and Permit Only PC6 & PC4



Allow:

- Network D (192.168.4.0)
- PC6 (192.168.3.2)
- PC4 (192.168.2.3)

```
enable
conf t
access-list 30 permit 192.168.4.0 0.0.0.255
access-list 30 permit host 192.168.3.2
access-list 30 permit host 192.168.2.3
access-list 30 deny any
interface Gig1/0
ip access-group 30 in
end
```

Output

PC10 From Network D Allowed to Network A

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=124
Reply from 192.168.1.2: bytes=32 time<1ms TTL=124
Reply from 192.168.1.2: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC6 Allowed to Network A

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=11ms TTL=125
Reply from 192.168.1.2: bytes=32 time<1ms TTL=125
Reply from 192.168.1.2: bytes=32 time<1ms TTL=125
Reply from 192.168.1.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 2ms
```

PC4 Allowed to Network A

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Discussion

1. ACLs were configured to control network traffic based on defined rules.
2. Traffic filtering was achieved using permit and deny statements.
3. Standard ACLs were applied based on source IP addresses.
4. Network security was enhanced by restricting unauthorized access.
5. ACL rules were processed sequentially for packet filtering.
6. End-to-end connectivity was verified using ICMP testing.
7. Controlled communication was successfully established between selected hosts.